Beyond Detection: AI-Powered Safety Violation Reporting for Construction Sites

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INTRODUCTION & PROBLEM

Introduction

We present an **AI-powered construction site safety monitoring system** that
automatically detects PPE violations and generates regulation-cited incident reports

Problem

- Current systems output only a detection label (e.g. "no hard hat") with **no contextual analysis**
- Supervisors must **manually identify violated regulations**, assess risk severity, and write incident reports
- **No automated link** between visual detection and regulatory compliance
- As construction sites scale, manual reporting **becomes increasingly time-consuming and error-prone**

Solution

A three-stage AI pipeline (Detect, Describe, Regulate) converts a construction site image into a fully cited WorkSafeBC incident report automatically.

METHODS — 3-STAGE PIPELINE

Construction site image

STAGE 1 Detect

PPE model	Zone model	Harness model
YOLOv8 · conf 0.3	YOLOv8 · conf 0.4	YOLOv8 · conf 0.3

OUTPUT TO STAGE 2

```
trigger_candidates: ["missing_high_vis"]
zone: "restricted-zone"  nearby_equipment: ["machinery"]
```

STAGE 2 Describe

Gemini 2.5 Flash — multimodal prompt

crop image + metadata (zone, equipment, PPE counts)

OUTPUT TO STAGE 3

```
violation_types: ["missing_high_vis"]  severity: "high"
description: "No high-vis vest, machinery overhead"
```

STAGE 3 Regulate

Rule-based selection	FAISS IndexFlatIP
violation_types · zone · equipment	cosine sim · score ≥ 0.5 · 21 chunks

TO LLAMA 3.3

```
citations: ["8.24", "8.25", "8.8"]  risk_score: 22 / 100
```

LLaMA 3.3-70b → Formal supervisor incident report

RAG PIPELINE ARCHITECTURE

VIOLATION DATA

```
violation types:No High-Vis Vest   zone:restricted-zone   nearby equipment:machinery
```

SELECT TOP 3 SIMILAR RULE

RULE DATABASE (21 WORKSAFEBC SAFETY RULES)

```
high vis vest+equip. → §8.24 §8.25
high vis vest → §8.25
hard hat+elevated → §8.11(1) §8.11(4)
harness+elevated → §11.4 §11.9
fallback(<3) → §8.8 §8.9
```

LLM INPUTS

Top 3 Similar Rule	JSON file	Risk Score	System Prompt
WorkSafeBC	violation data	base × zone × severity	

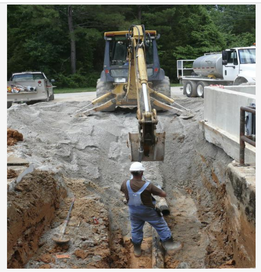
LLM
LLaMA 3.3-70b

Formal supervisor incident report

Summary: Stage 3 queries a Rule Database of 21 WorkSafeBC regulations, selects the Top 3 most similar rules, and passes them alongside the violation data, Risk Score, and system prompt to LLaMA 3.3-70b to generate a formal incident report.

RESULTS — EXAMPLE OUTPUT

worker_001 · restricted-zone | **Risk 22 / 100**



VIOLATIONS

Missing High-Vis Vest

SEVERITY

HIGH — active machinery overhead in trench

CITED REGULATIONS

§8.8 §8.24 §8.25

SUPERVISOR REPORT

- Date / Time: 2026-04-07 · 13:35
- Location: Restricted trench zone
- Observation: No high-vis apparel, active machinery overhead
- Violation: WorkSafeBC §8.24, §8.25
- Action: Immediate corrective action required

APPLICABLE REGULATIONS

§8.24 Worker exposed to vehicles over 30 km/h must wear Class 2 or 3 high-vis apparel (CSA Z96-15).

§8.25 Apparel must contrast with environment and include 775 sq cm fluorescent/retroreflective trim.

§8.8 Supervisor must ensure PPE is available, properly worn, cleaned, inspected and stored.

Key insight: combining rule-based priority selection with semantic similarity retrieval eliminates hallucinated citations while maintaining full recall across all violation types.

RISK SCORE = BASE SCORE OF VIOLATION TYPE × ZONE × SEVERITY

Violation Type	Base Score	Zone ×	Severity ×	Max
Missing harness	25	×2.0 (elevated)	×1.5 (high)	75
Missing hard hat	15	×1.5 (restricted)	×1.5 (high)	34
Missing high-vis vest	10	×1.5 (vehicle)	×1.5 (high)	23

Base Score — Predefined risk score per violation type:
Missing harness = 25, Missing hard hat = 15,
Missing high-vis vest = 10

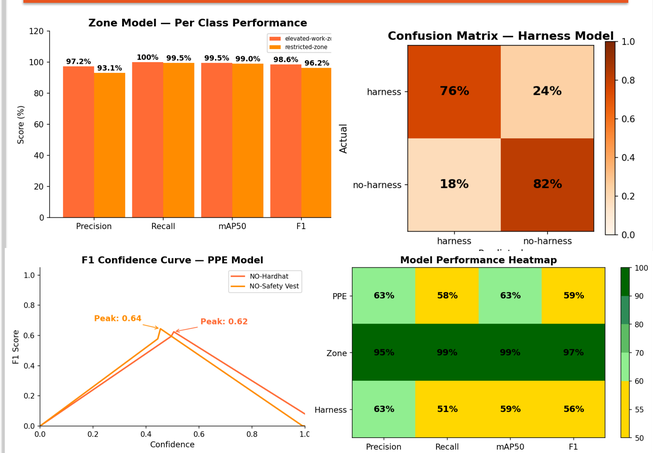
Zone Multiplier — Risk weight by work zone:
elevated-work-zone = ×2.0, restricted-zone = ×1.5,
vehicle-zone = ×1.5, safe = ×1.0

Severity — Contextual severity assessed by Gemini from crop image:
high = ×1.5, medium = ×1.0, low = ×0.5

Max Score — Maximum possible score per violation type under worst-case zone and severity conditions

Reference: Base scores reflect risk severity and obligation strength as described in WorkSafeBC OHS Regulation.

MODEL PERFORMANCE



Zone detection is highly accurate, **PPE violation detection** is at a satisfactory level, while **harness detection** still has room for improvement.

SYSTEM STACK

Detection:	YOLOv8
VLM:	Gemini 2.5 Flash (Google GenAI SDK)
Embeddings:	sentence-transformers all-MiniLM-L6-v2
Vector store:	FAISS IndexFlatIP
LLM:	LLaMA 3.3-70b-versatile
UI:	Streamlit
Regulations:	WorkSafeBC OHS Regulation 2026

CONCLUSIONS

- Three-stage pipeline transforms bare detection labels into fully cited formal incident reports — zero hallucinated regulations.
- Zone-aware and equipment-aware RAG outperforms pure semantic search for regulatory retrieval.
- From-scratch FAISS pipeline is lightweight and interpretable.
- Modular design: adding a new violation type requires only a new YOLO label + WorkSafeBC chunk.

FUTURE WORK

- Expand to mask, gloves, safety-shoe detection (Keremberke dataset).
- Heavy-equipment proximity detection (excavator/dump truck IoU).
- Real-time video with ByteTrack worker tracking.
- Multi-language report generation for diverse work sites.
- Automated alert dispatch to supervisors.

REFERENCES

[1] WorkSafeBC, "OHS Regulation," Jan. 2026
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[6] Zhang et al., "ByteTrack," ECCV, 2022.

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